TABLE I

NITTO DENKO CORP.—MANUFACTURING LINE FOR ADHESIVE TAPES FOR ELECTRONICS

Cost classification	Material	Energy	System	Waste management	Total
Product	¥2,499,944 (68.29%)	¥57,354 (68.29%)	¥480,200 (68.29%)	—	¥3,037,498 (67.17%)
Material loss	¥1,160,830 (31.71%)	¥26,632 (31.71%)	¥222,978 (31.71%)	¥74,030 (100%)	¥1,484,470 (32.83%)
Total	¥3,660,774 (100%)	¥83,986 (100%)	¥703,178 (100%)	¥74,030 (100%)	¥4,521,968 (100%)

TABLE II

SUMIRON CO. LTD—SMALL-TO-MEDIUM BUSINESS AND MASS PRODUCTION

Cost classification	Material cost	Energy cost	System cost	Waste management cost	Total
Product	40,300,000 (53.3%)	2,700,000 (3.6%)	8,900,000 (11.8%)	—	51,900,000 (68.7%)
Material loss	16,600,000 (22.0%)	1,600,000 (2.1%)	5,400,000 (7.1%)	—	23,600,000 (31.2%)
Disposed of/recycled	—	—	—	90,000 (0.1%)	90,000 (0.1%)
Sub-total	56,900,000 (75.3%)	4,300,000 (5.7%)	14,300,000 (18.9%)	90,000 (0.1%)	75,590,000 (100.0%)

lowed in product designing and implementation

The development, quality assurance and quality control team is mostly responsible for reducing the costs of quality and non-quality. Research says that most recurrences from the field that get consumers unhappy are technical issues, including ones due to behaviour of a product with other connected devices, reliability, stability, performance and so on. This requires skilled employees (both proof and evolution) who can identify issues related to design, protocol or codification.

Owing to the immense pressure to establish the product early in the market, organisations focus more on getting the work done, bypassing many quality-related processes (which increase the software deliverable process time), which may also impact the fast pace of field returns. By this point, it is too late to contain the cost of non-quality.

The challenge is to manage this state of affairs, so that industries can optimise the overall monetary value without compromising the quality of products. Below are a few factors that may allow organisations to achieve such goals.

Of the many reasons for the cost of non-quality, a few important ones are explained below.

Cost of rework. Repeated tasks to achieve a certain goal is called rework. This may happen if engineers have not performed error-free work at the first try. Rework is the most common problem that many organisations face in their day-to-day project lifecycle. Consequently, situations of schedule constraints create extreme pressure for the development team, which may result in the lack of involvement in the testing phase.

Below are some probable points due to which rework may occur:

- Faults not solved correctly, non-conforming with regards to coding standards or logic
- Very frequent change in requirements, design, etc, during development and testing stages
- Incomplete coverage in code implementation or hardware implementation during design implementation phase
- Test executions or design implementations not measured or evaluated; hence, incomplete coverage
- Incorrect test plan and test strategy creation
- Frequent regressions
- Negligence by developers or test engineers

Cost of frequent design/requirement change. Design changes in software are very common during the software development lifecycle. Generally, design changes come in when projects are in the middle of software development lifecycle, which contribute to a lot of rework. Below are a few points that indicate the impact of design changes:

> "A five per cent reduction in defect rate can increase profits by five to 95 per cent."
>
> —***Bain & Company***

- Design changes in one component often impact other components, which often encourages software changes in other components as well and, finally, rework in one or more software elements
- Increasing overall development cost
- High chance of delay in the market
- High rate of regressions
- May need to develop additional components and, hence, boost efforts with respect to planned efforts

To avoid these issues, it may be useful for organisations to ask a few questions during conception phase and create a secure database that may provide optimal information to recover the shock of design changes.

Some such questions that should